

Predicting Near-Term Customer Value to Improve Promotional Targeting

by

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1. Introduction:

Everyday our mailboxes and inboxes are flooded with promotions from businesses hoping to maximize revenue. However, not every promotion leads directly to future revenue. One can assume many will be ignored. How, then, can businesses identify which customers are most valuable to engage right now?

Most retail businesses have a finite marketing budget. A poorly targeted promotional campaign can significantly reduce return on investment. If a company could identify those customers likely to generate the most income in the near-future, said company could focus promotions, discounts, and engagement efforts more effectively. Marketing teams, customer relations management personnel, and business analysts responsible for improving campaign performance stand to benefit from this business insight.

This project utilizes the transactional retail data from the [UCI Machine Learning Repository](#) in order to build a predictive model that estimates customer revenue over the next 60 days. The goal is to help businesses prioritize customers for targeted promotions and maximize near-term customer value.

2. Dataset:

The dataset used in this project comes from the [UCI Machine Learning Repository Online Retail](#) dataset, which contains transactional records from a UK-based online retailer. The original data set included information such as: Invoice Number, Product Description, Quantity Purchase, Unit Price, Customer ID, and Transaction Date. The data was downloaded directly from the repository and processed using Python for analysis and modeling.

Before generating customer-level features, the data had to be cleaned and manipulated. All canceled transactions, as well as missing Customer IDs, were removed. A 'Revenue' column was generated by multiplying 'Unit Price' by 'Quantity.' These steps ensured that the dataset was clean, consistent and appropriate for modeling and near customer value.

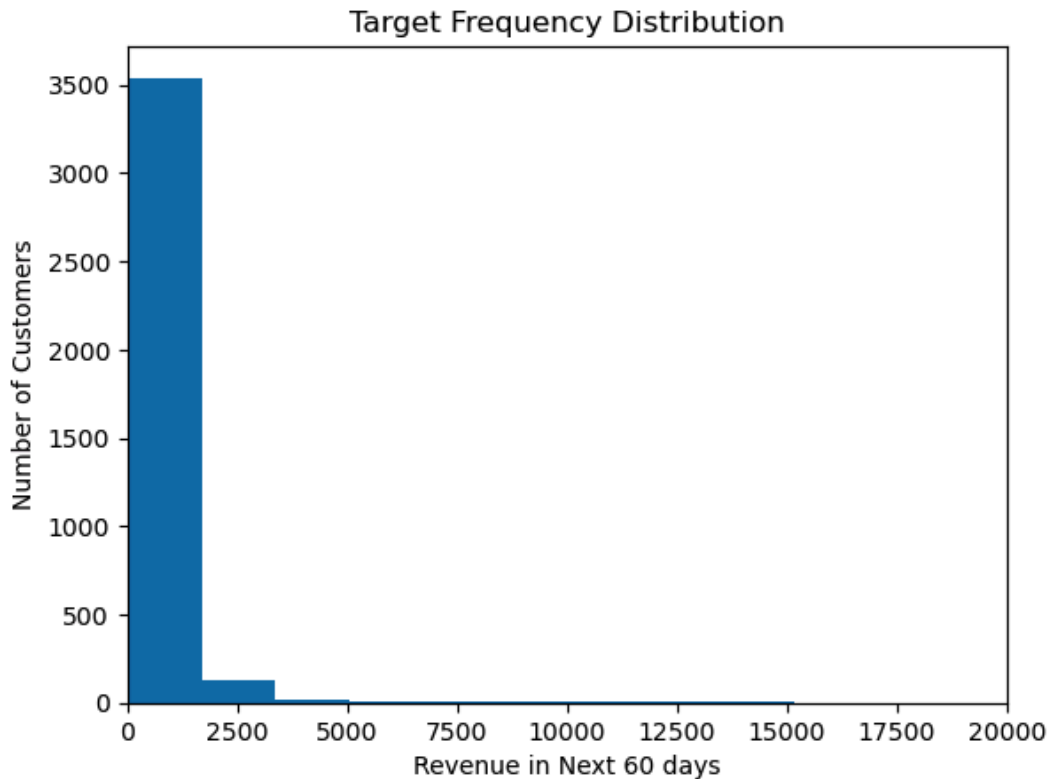
By aggregating these transaction-level features into customer-level features, each row could represent a unique customer at a specific snapshot in time. A cutoff date (2011-10-10) was selected in order to see what each customer looked like before the cutoff and then observe how much revenue they generated over the next 60 day period. This Customer revenue in the next 60 days defined the target variable. Prior to the cutoff date, Customer-level features were calculated. These include: Recency (days since last purchase), Frequency (number of purchases), Monetary Value (total historic spending), and Average Order Value (a behavioral proxy calculated by dividing Monetary by Frequency).

In the end, my final dataset had one row per customer with that customer's respective 'Recency', 'Frequency', 'Monetary Value', and 'AOV' as feature columns, as well as the target variable '60_day_revenue.'

3. Exploratory Data Analysis:

To better understand the dataset and identify patterns in customer behavior, several visualizations were created.

A. Univariate Analysis: Distribution of 60-Day Revenue



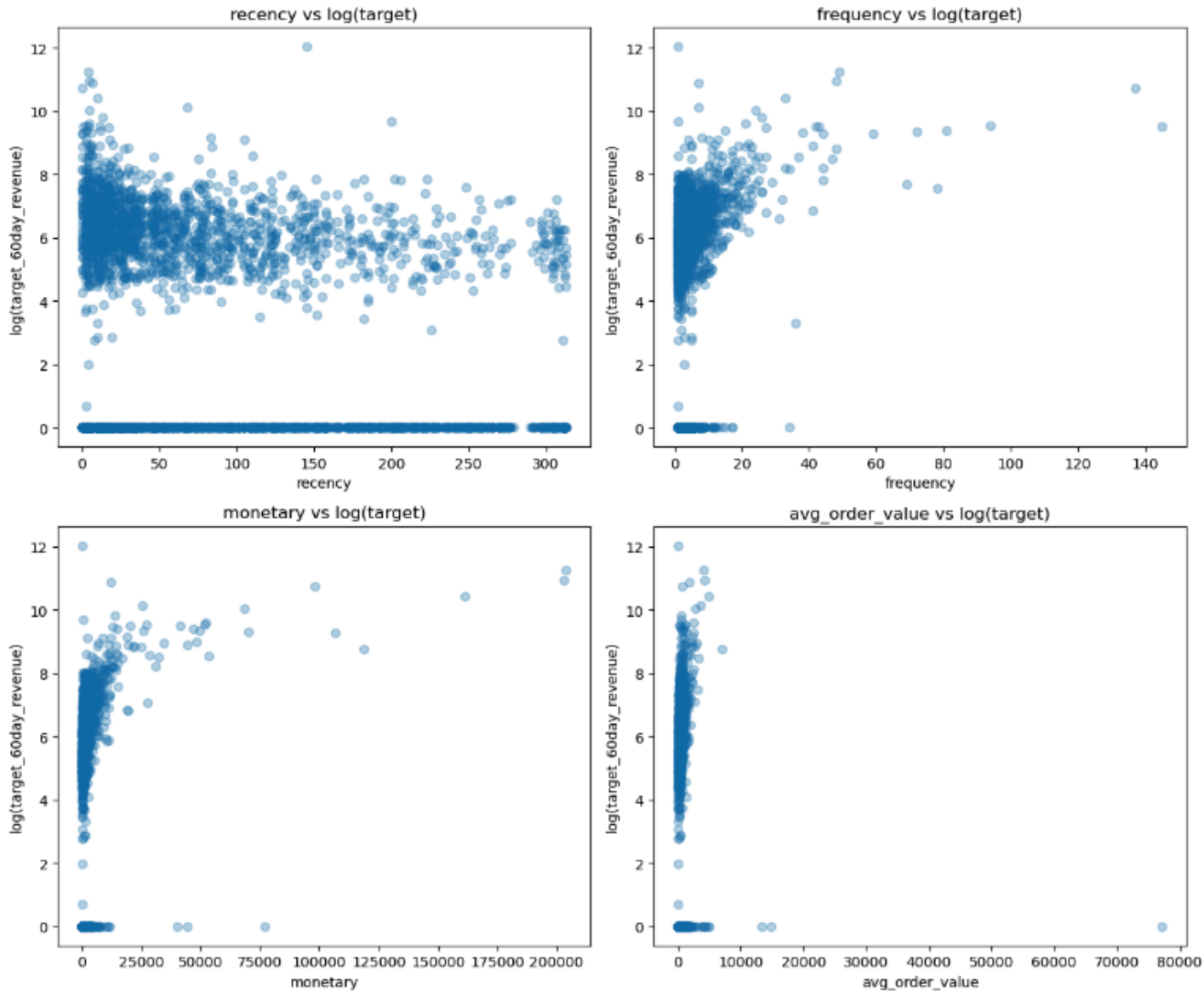
This chart shows how future customer revenue is distributed across the dataset. There is a right-skew with a long tail. We can infer that many customers are clustered at very low values close to or at zero. Most customers buy very little. This is known as zero-inflation. As the tail stretches rightwards, we start to see a small number of customers who have large revenue values.

This confirms the importance of predicting high-value customers accurately.

Note: Due to the high-skew of the target, it was Log Transformed for the rest of the project in order to better visualize relationships.

B. Bivariate Analysis-

I. Scatterplots

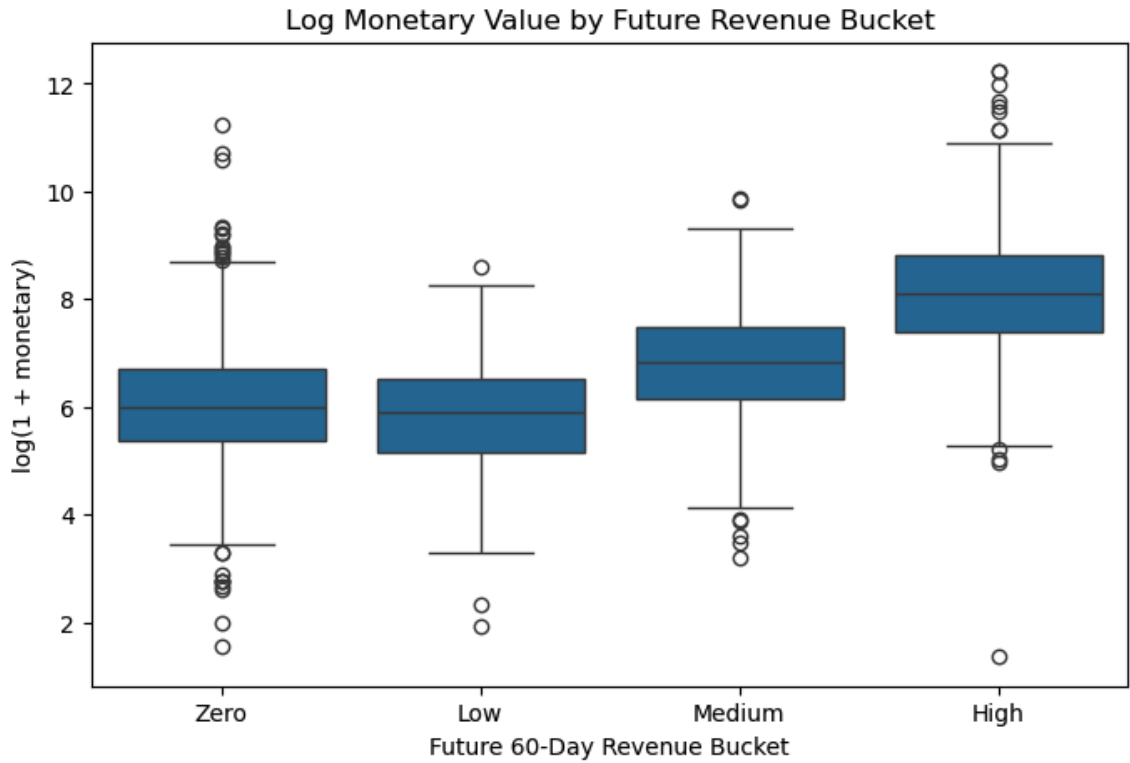


From the above scatterplots, one can see how each feature has a weak-to-moderate visible relationship with the 60-day revenue target. 'Frequency' and 'Monetary' show somewhat positive trends, while 'Recency' appears non-linear. Due to heavy skew and zero-inflation in the target variable, additional analyses will be used to better assess feature relationships.

II. Bucketed Boxplots

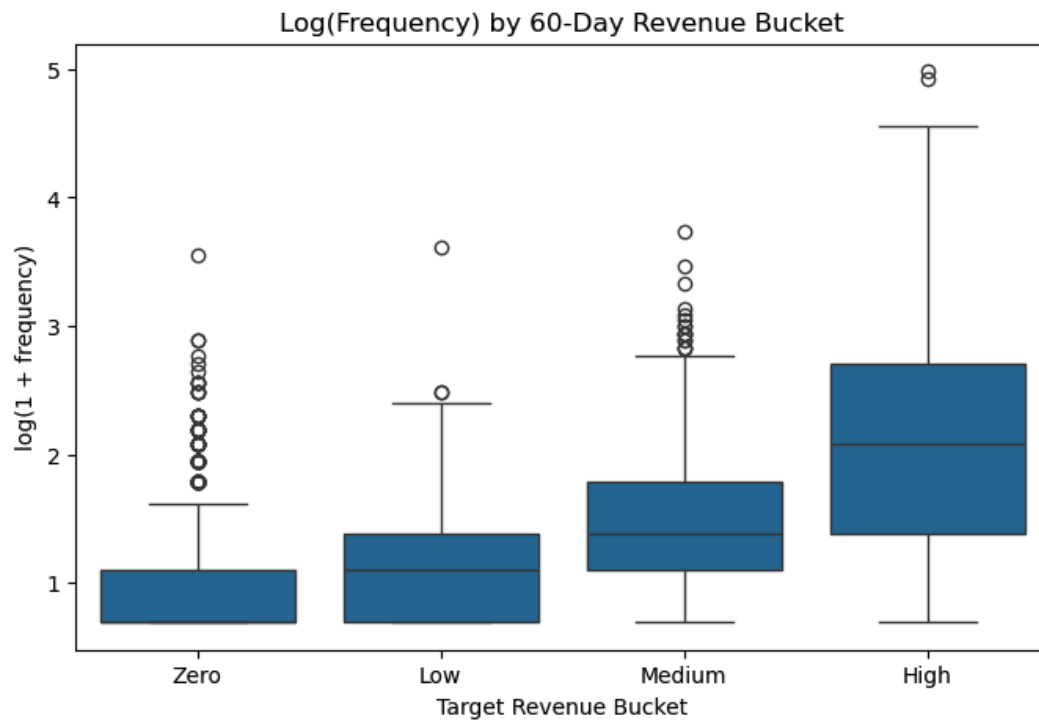
The target variable was bucketed into groups in order to observe if higher target buckets show higher 'Recency', 'Frequency', 'Monetary value', or 'Average Order Value.'

A. Monetary Value vs Future Revenue:



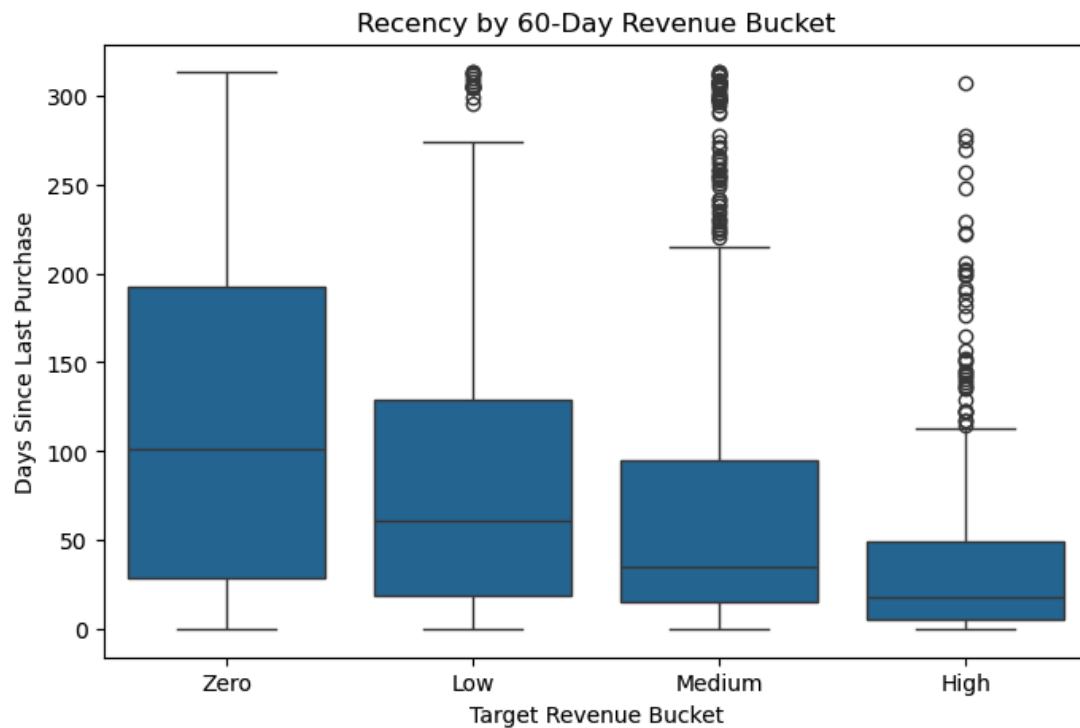
'Monetary' value increases across higher-future revenue buckets. However, the data shows the "Zero" revenue in the future group to be slightly bigger spenders in the past than the "Low" revenue in the future group. This suggests that while prior spending is predictive, not all historically strong customers repurchase within the 60-day window.

B. Frequency vs Future Revenue



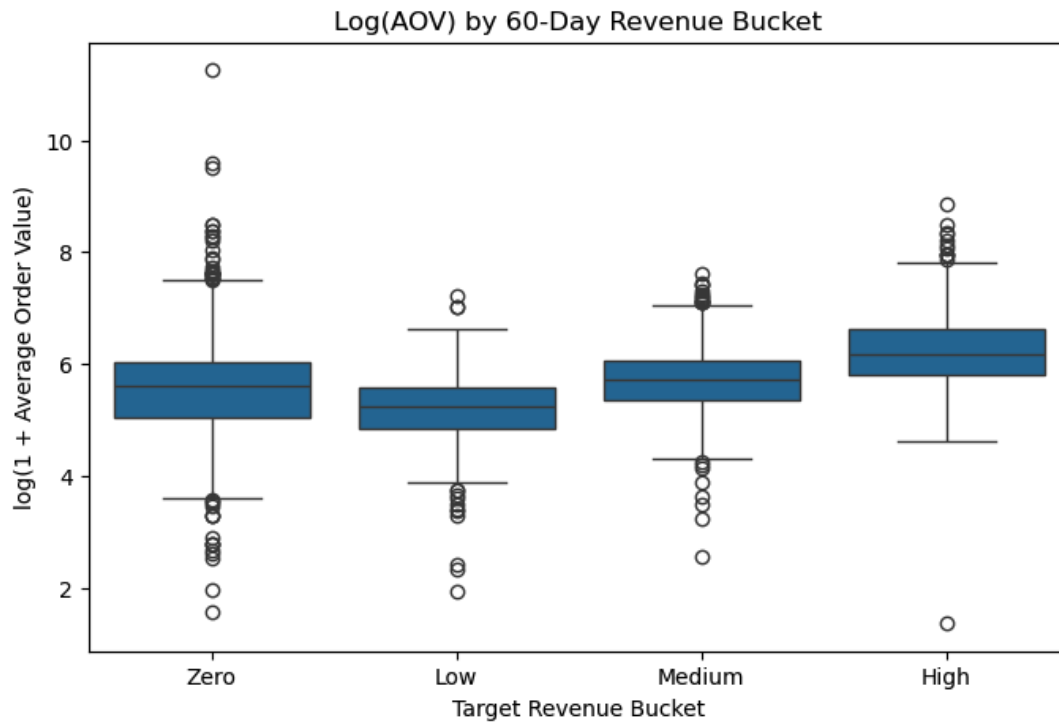
The 'Frequency' median ordering is as follows Zero<Low<Medium<High. This means that customers who bought more often before the cutoff window are likely to spend more in the next 60 days. This chart helps confirm purchase Frequency is predictive of future spending.

C. Recency vs Future Revenue



From the graph, one can see the 'Recency' medians appear to descend from 'Zero' to 'High'. In other words, customers in the high target revenue bucket have the lowest median days since last purchase, while the customers with zero revenue have the highest median. In other words, customers who bought more recently before the cutoff will likely buy again soon. This relationship is consistent with marketing theory and customer lifecycle behavior.

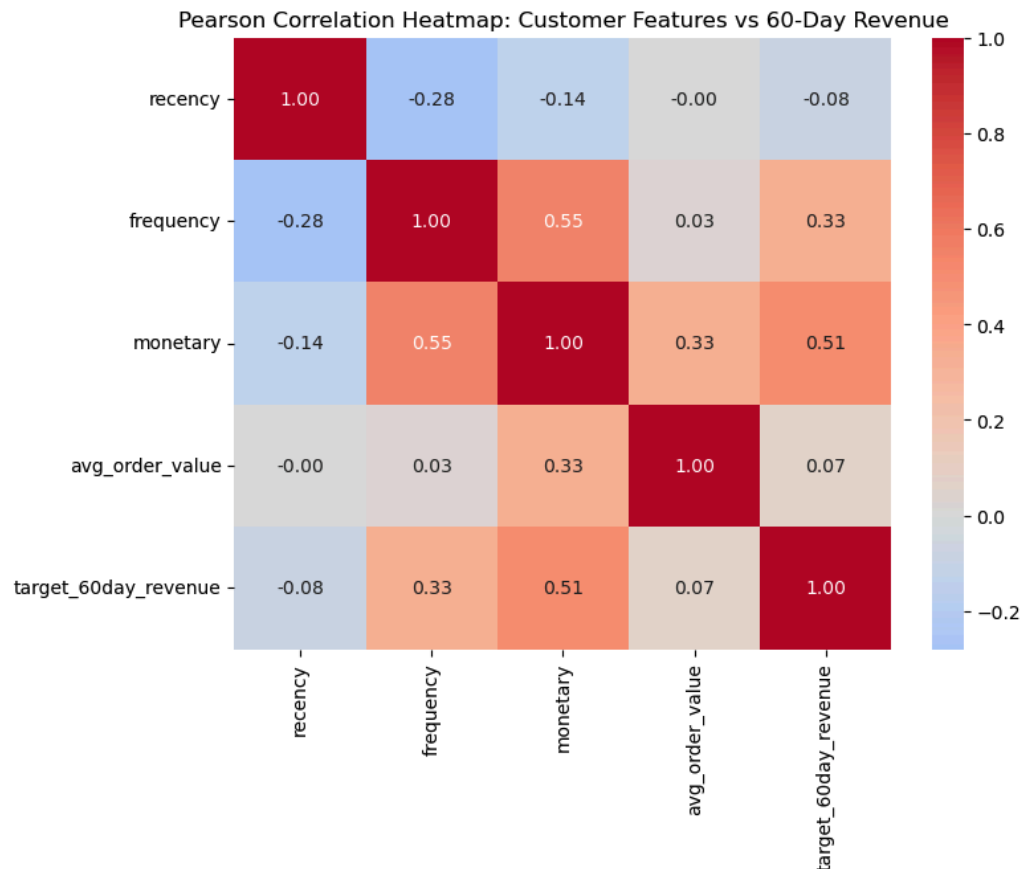
D. Average Order Value vs Future Revenue



Similar to the monetary plot, the median historical 'Average Order Value' order ascends over target revenue low<zero<medium<high. This means some customers in the zero bucket (those who made no purchases in the 60 day window) still had historically high AOV. That said, medium and high buckets show the expected increase in median. One can conclude that AOV is a weak signal, but still useful as a proxy if combined with other meaningful features such as frequency and recency. Alone, however, it may mislead.

C. Correlation Matrix

A correlation matrix was created in order to visually and mathematically visualize relationships between features, potential multicollinearity, and features strongly related to the target variable.



For each bivariate relationship explored above, the Pearson correlation test was used to determine whether the observed relationships were statistically significant.

From the heatmap one can see that 'Frequency' and 'Monetary' have the largest positive correlations, with 'Monetary' being the strongest linear relationship. AOV is a positive correlation but much weaker. 'Recency' has a negative linear correlation. From the features themselves, AOV appears positively correlated with 'Monetary'. Also 'Frequency' is positively correlated with 'Monetary'.

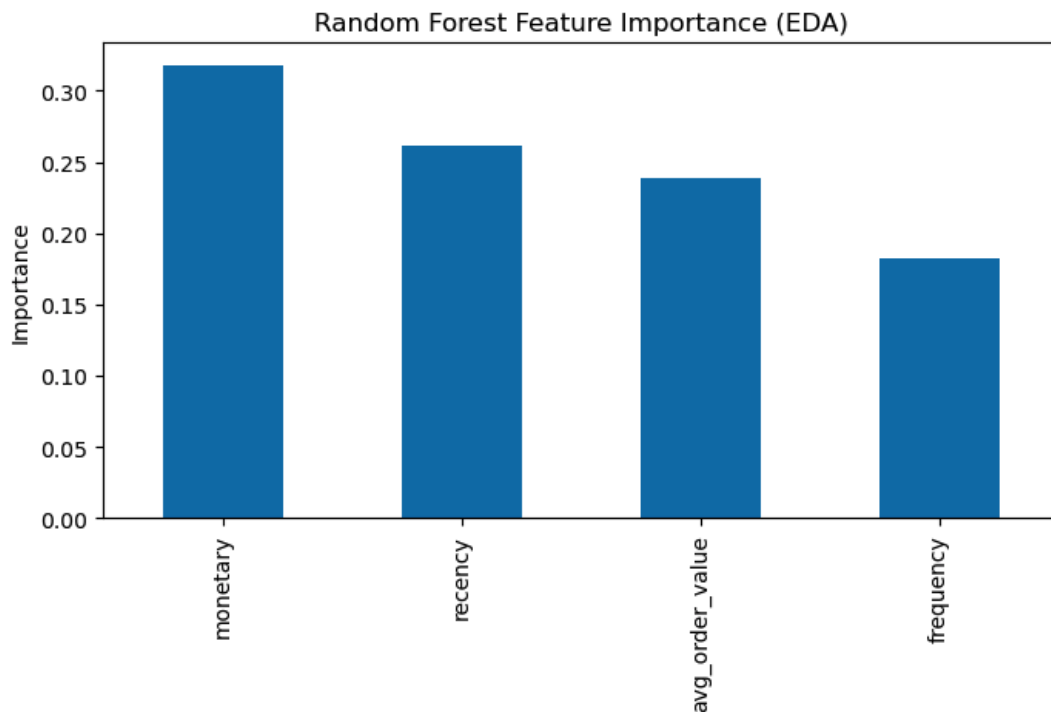
No extreme multicollinearity was observed that would impact model stability.

Thus, customers who buy more will likely buy again soon (frequency). Customers who historically spend more tend to generate more near-term revenue. High AOV won't guarantee

near-term revenue. Lastly, very recent customers behave differently, but after a certain point recency will stop being predictive.

D. Feature Importance

Exploratory feature importance was performed via Random Forest to see whether or not a non-linear model would agree with the above linear insights.



From the graph, one can see 'Monetary' and 'Recency' being the top non-linear drivers. Recency's ascendance in importance is due to its non-linear nature. Like 'Recency', AOV looked different in correlation than it does in Random Forest. We can see now that AOV has a predictive relationship to near-term value. Thus, these two features were retained in modeling.

4. Modeling

To identify the best predictive approach, three models were tested: Ridge Regression, Random Forest, and Gradient Boosting. Each model was hyperparameter-tuned using GridSearchCV and evaluated using RMSE.

Model Performance Comparison

Model	Key Parameters	RMSE	RMSE (£)
Ridge Regression	alpha=1 (default)	2.7620	14.831474
Random Forest (tuned)	n_estimators=300, max_depth=5, min_samples_split=10, min_samples_leaf=4	2.7586	14.777739
Gradient Boosting (tuned)	n_estimators=300, learning_rate=0.01, max_depth=3, min_samples_split=2, min_samples_leaf=4	2.7573	14.757241

These three models performed very similarly; tree-based models gave a tiny improvement over Ridge.

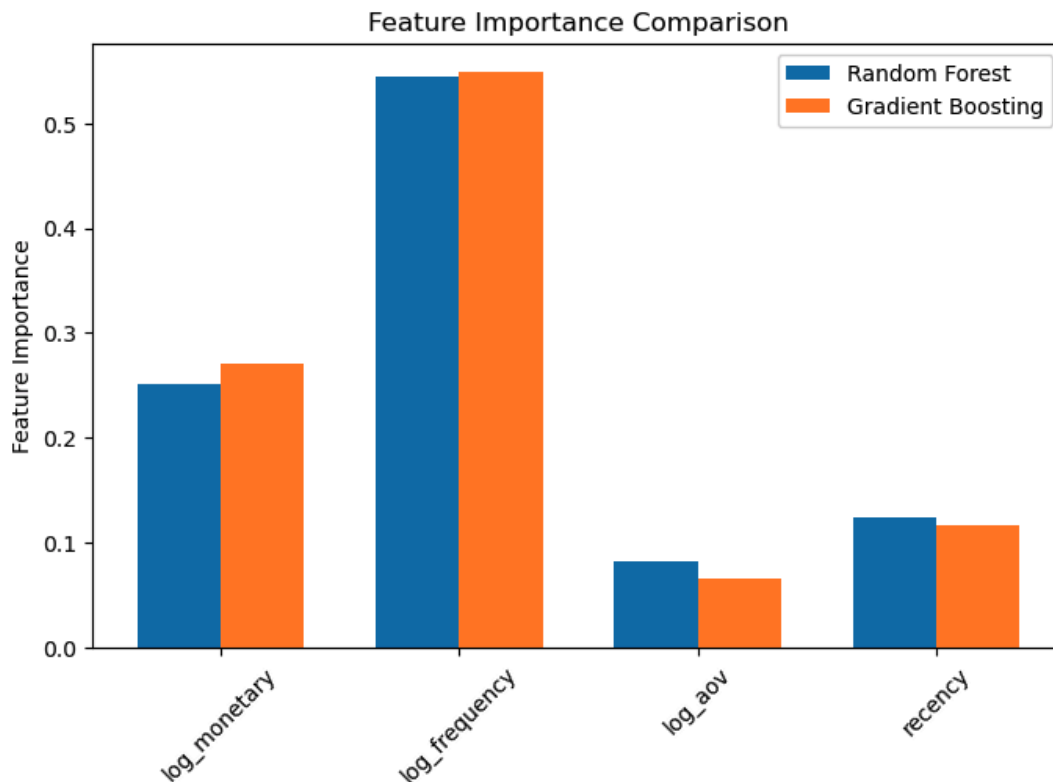
Gradient Boosting achieved the lowest RMSE. Due to its slightly stronger predictive performance and ability to capture non-linear relationships, Gradient Boosting was selected as the final model.

In business terms, the model's predictions will be off by approx £14.75 per customer.

Feature Importance Comparison

After Gradient Boosting was selected as the best model, feature importance was performed in order to discern which features best drive prediction. This allows the business to better understand their ideal high-value customer.

Random Forest was also included in feature importance in order to see if there were any discernable differences in feature preference.

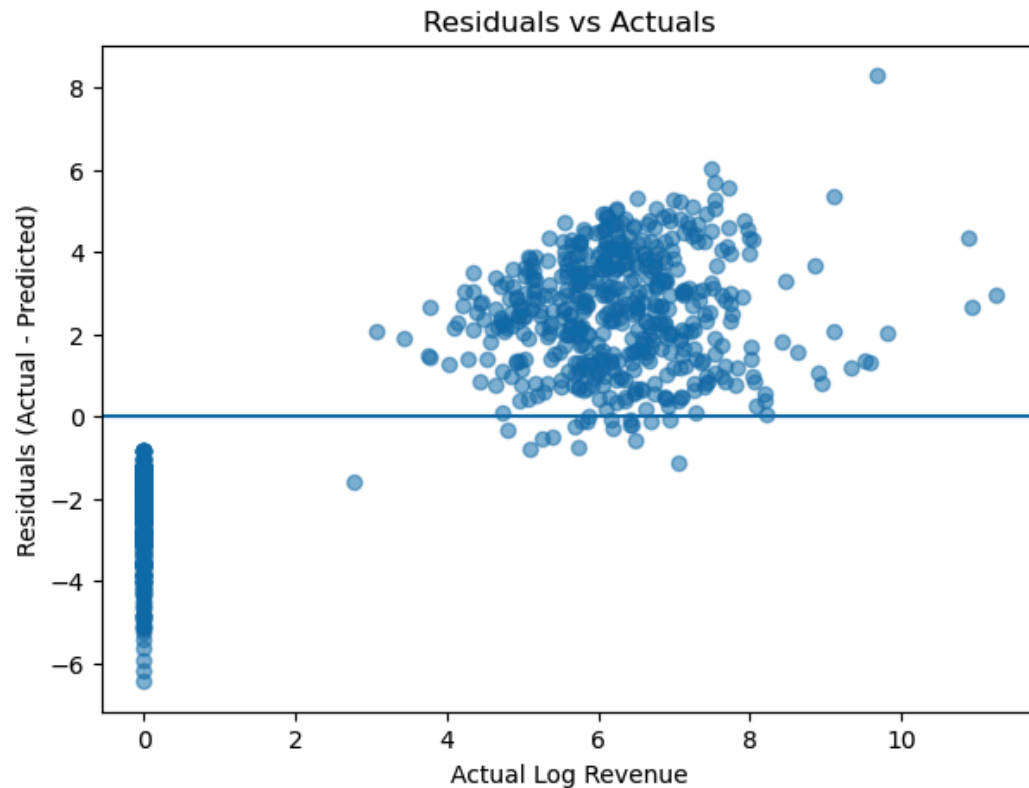


Both models found the exact same thing: 'Frequency' is the dominant driver of predictions.

The most valuable customers in the next 60 days are those who purchased most often. This makes sense because frequent buyers are active, engaged, and likely to return. Thus, **'Frequency' is the strongest predictor of future revenue.**

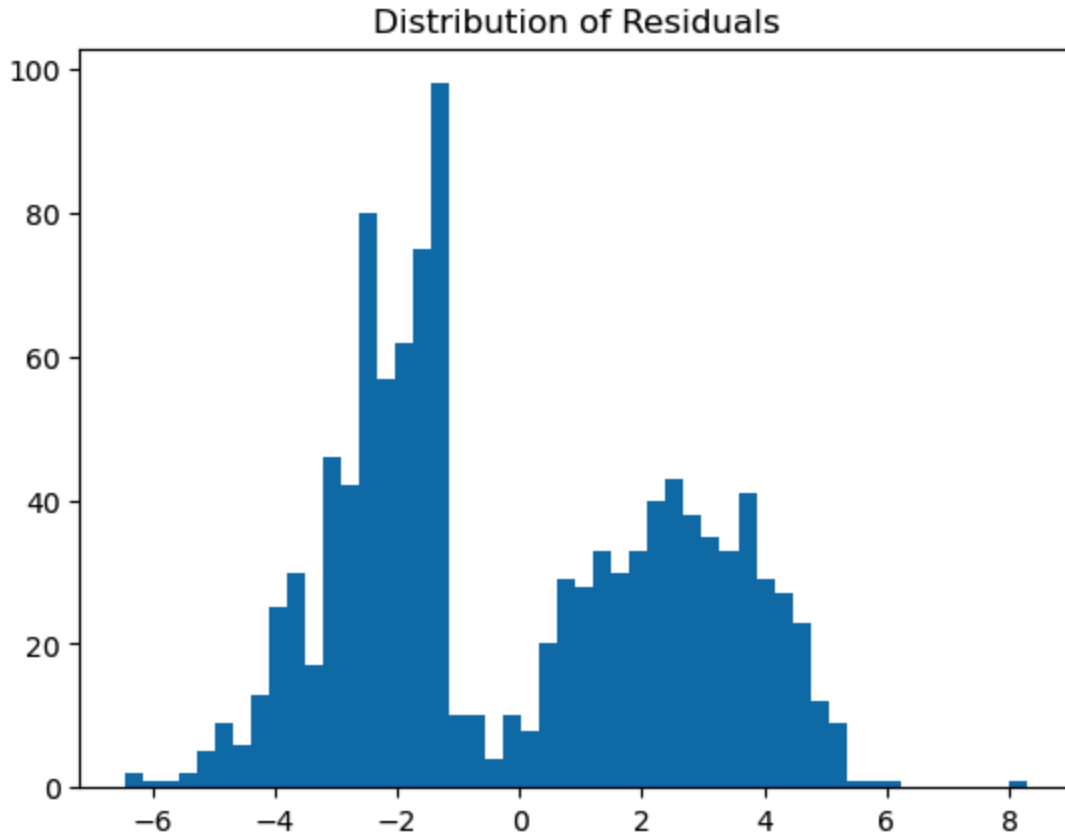
Residual Analysis

Residual Analysis was performed in order to check for model bias, underprediction of high-value customers, and/or overprediction of low-value customers.



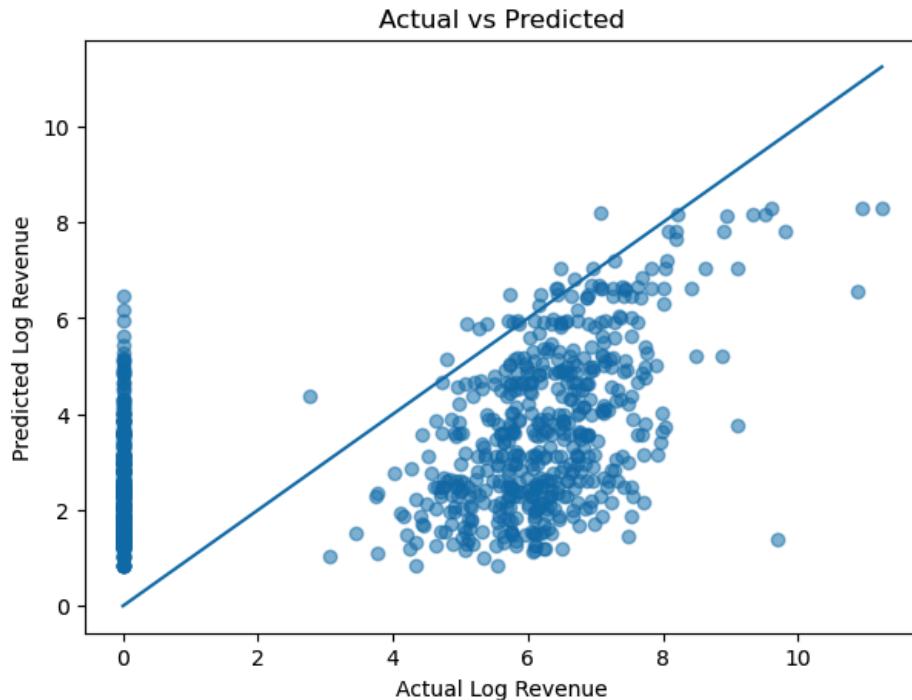
Residual analysis shows that prediction errors are generally centered close to zero overall, with a slightly negative median residual. This indicates the model tends to modestly overpredict revenue for many customers, particularly those who ultimately generated little or no revenue during the 60-day window. This behavior is expected in datasets with a large proportion of zero-revenue customers, where models may predict small amounts of revenue for inactive customers. Despite this, the overall distribution of residuals suggests the model does not exhibit strong systematic bias.

This confirms what was seen in the EDA. Namely, that the dataset is zero-heavy.



The residual distribution shows two distinct clusters, reflecting the presence of two different customer behaviors within the dataset. The taller cluster corresponds to customers who generated no revenue during the prediction window, where the model tends to slightly overpredict small amounts of revenue. The second cluster represents customers who made purchases where prediction errors are more widely distributed. This pattern highlights the challenge of modeling customer revenue in datasets with a high proportion of zero outcomes.

The key take away is the model slightly over-predicts inactive customers and slightly under-predicts active customers. This is what creates the "two-lump" distribution.



The model demonstrates a strong positive relationship between actual and predicted revenue. However, predictions for high-revenue customers tend to fall below the ideal prediction line, indicating that the model slightly underestimates customers with very high future spending. This may be due to the skewed distribution of spending and the rarity of extreme values. This confirms previous findings in residual analysis.

5. Conclusion:

This project demonstrates how transactional retail data can be used to identify customers who are most likely to generate revenue in the near future. Using the **UCI Machine Learning Repository Online Retail dataset**, multiple machine learning models were evaluated to predict customer revenue over the next 60 days. Among the models tested, **Gradient Boosting achieved the lowest RMSE**, making it the most effective model for predicting near-term customer revenue with reasonable accuracy.

Feature importance analysis revealed that **'Frequency'** was the most influential predictor of future revenue. In other words, customers who purchase more frequently are significantly more likely to generate revenue in the near future. This aligns with expectations from marketing analytics, as frequent purchasers tend to exhibit stronger engagement and loyalty. Additionally, **historical spending ('Monetary' Value)** and **'Recency'** also played important roles in predicting future revenue, indicating that customers who have spent more in the past and have purchased recently remain valuable targets for marketing efforts.

These insights can help businesses target promotions more effectively and efficiently. By focusing on customers who are most likely to generate revenue in the near term, marketing teams can reduce wasted resources on low-value prospects and improve the overall return on investment of their campaigns. Ultimately, this approach enables marketing teams to shift from broad promotional campaigns to data-driven customer prioritization.

Based on these findings, several concrete recommendations can be made.

First, the business should prioritize high predicted revenue customers for promotions. Customers can be ranked based on their predicted revenue over the next 60 days, allowing marketing teams to focus on the top 10–20% of high-value customers. These individuals should receive personalized promotions, early access to new products, or loyalty incentives designed to encourage continued engagement and spending.

Second, the company should re-engage recently inactive but historically valuable customers. The exploratory analysis and feature importance results indicate that 'Recency' and 'Monetary' value strongly influence future spending. Customers who previously spent significant amounts but have not made a recent purchase represent an opportunity for targeted win-back campaigns. Offering personalized discounts, reminders, or tailored marketing messages could encourage these customers to return.

Third, the predictive model should be integrated into ongoing marketing strategy and decision-making. Rather than using this analysis as a one-time exercise, the company can update predictions regularly and incorporate them into customer relationship management and campaign planning processes. Doing so would allow the business to allocate marketing resources more effectively and continuously improve targeting over time.

For next steps and future work, several improvements could further enhance this analysis. Additional customer features, such as product category preferences or geographic information, could provide deeper insight into purchasing behavior. More advanced models, such as XGBoost, could also be tested to determine whether predictive performance can be improved.

Residual analysis also revealed two distinct clusters of prediction errors, reflecting the presence of both purchasing and non-purchasing customers within the dataset. Future work could address this by implementing a two-stage modeling approach, where one model predicts the likelihood of a customer making a purchase and a second model predicts the expected revenue for purchasing customers. This approach may improve prediction accuracy for both customer groups.

In addition, future analysis could evaluate the effectiveness of marketing campaigns that use predicted customer value, helping validate the business impact of the model. Finally, combining customer segmentation with revenue prediction may provide an even more powerful approach to optimizing marketing strategy.

Overall, this project demonstrates how predictive analytics can transform raw transactional data into actionable insights that help businesses identify, prioritize, and engage their most valuable customers.